List of Figures

List of Tables

1 Introduction

Inflation prediction is crucial for policymakers, businesses, and consumers as it influences decisions on interest rates, investment strategies, and wages. Accurate forecasts help central banks set appropriate monetary policies to maintain economic stability and control price levels. Businesses rely on inflation expectations to plan budgets, pricing strategies, and resource allocation, while consumers consider inflation trends when making purchasing and savings decisions. However, inflation prediction is challenging due to the interplay of numerous dynamic factors that influence prices, such as monetary policy, supply chain disruptions, labor market conditions, and geopolitical events. Predicting inflation components, such as food, gas, and clothing, adds another layer of complexity, as these categories are affected by distinct factors like weather, global markets, and trade policies. Each component has unique drivers, making it difficult to aggregate these predictions into a comprehensive forecast. Additionally, inflation expectations create feedback loops—when businesses and consumers anticipate rising prices, their actions, like increasing wages or raising prices, can directly contribute to higher inflation. These factors, combined with global and local economic interactions, make accurate inflation prediction a significant challenge.

The Consumer Price Index (CPI) is organized hierarchically, grouping goods and services into broad categories such as food, energy, and housing, which are further subdivided into detailed subcategories to capture their impact on overall inflation. This complex structure necessitates advanced modeling techniques. There are several methods for training predictive models with hierarchical data. One approach is to train separate models for each series within the hierarchy, which can reduce the risk of under-fitting by focusing on specific data segments. However, this method often leads to overfitting due to the limited amount of data available for each individual model. An alternative approach is to train a single model using all the hierarchical data combined, which can take advantage of larger datasets but tends to be computationally intensive and may struggle to capture the differing dynamics across various levels of the hierarchy. Our approach, BiHRNN, achieves an effective middle ground by harnessing hierarchical relationships to enhance model performance, without the computational burden and inefficiency of training a single, unified model.

Bidirectional Hierarchical Recurrent Neural Networks (BiHRNN) are designed to